Module 1

FUNCTIONAL MATERIALS FOR MEMORY AND DISPLAY SYSTEMS

Memory Devices: Introduction, organic semiconductors; types of organic semiconductors used in memory devices, p-type semiconductor-pentacene and n-type semiconductor-perfluoropentacene. Differences between organic and inorganic memory devices. Construction, working and advantages of pentacene semiconductor chip.

Resistive RAM (ReRAM) Materials: Introduction, synthesis of TiO_2-RAM nanomaterial by sol-gel method, properties and applications.

Display Systems: Introduction, liquid crystals (LCs)-classification, properties and its applications. Construction, working principle and applications of Light Emitting Diodes (LEDs), Organic Light Emitting Diodes (OLEDs), Active-Matrix Organic Light Emitting Diodes (AMOLEDs) and Quantum Light Emitting Diodes (QLEDs).

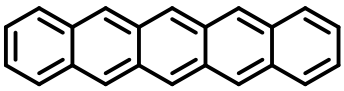
Semiconductors

Semiconductors are materials whose electrical conductivity lies between that of conductors and insulators. Their conductivity can be modified by temperature, light, or impurity doping, making them essential in electronic and optoelectronic devices such as diodes, transistors, solar cells, and LEDs.

In semiconductors, the valence band is partially filled or separated from the conduction band by a small band gap (0.1–3 eV).

p-type Semiconductor

Materials in which electrical conduction occurs mainly through holes (positive charge carriers) created due to electron deficiency are known as p-type semiconductors. These materials are typically formed either by doping with trivalent elements or by using compounds that possess high HOMO (Highest Occupied Molecular Orbital) energy levels, such as pentacene. Pentacene, an organic π-conjugated molecule, exhibits hole-dominated conduction and therefore functions as a p-type organic semiconductor.



Pentacene

Properties:

- Pentacene shows hole-dominated conduction through hopping between HOMO levels.
- It has HOMO ≈ −5.0 eV, LUMO ≈ −3.0 eV, and a band gap of about 2.2 eV, making it suitable for hole injection.
- The molecule is planar and π-conjugated, allowing strong π–π stacking, which enables efficient hole transport.
- It exhibits strong absorption in the visible region around 660 nm.

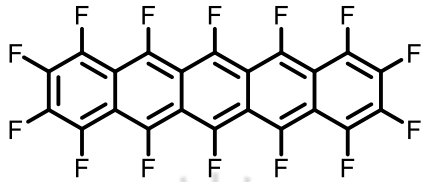
- Pentacene is sensitive to oxygen, moisture, and light, so it requires encapsulation for stable device operation.

Applications of pentacene:

- Organic field-effect transistor (OFET).
- Flexible and wearable electronics.
- Organic semiconducting chip.

n-type Semiconductor

Perfluoropentacene (PFP) functions as an n-type organic semiconductor, a material used in organic electronics due to its ability to conduct electrons. It is derived from the p-type organic semiconductor pentacene by fluorination.



Perfluoropentacene

Properties:

- It has a high electron affinity, which is important for its behavior as an n-type semiconductor.
- The material shows strong π–π stacking, allowing efficient charge transport in thin films.
- It exhibits electron mobility in organic field-effect transistors (OFETs), with values around 0.11 $cm^2/V \cdot s$ observed in some devices.

Applications:

- It is used as an n-type material in high-performance organic field-effect transistors (OFETs) and ambipolar transistors, which can operate at both positive and negative gate voltages.
- It works together with pentacene (a p-type material) to form complementary inverter circuits or p–n junctions, showing high voltage gain.
- Its electronic properties are also applied in organic non-volatile memory devices, where it helps in storing data efficiently.

Difference between organic and inorganic semiconductors

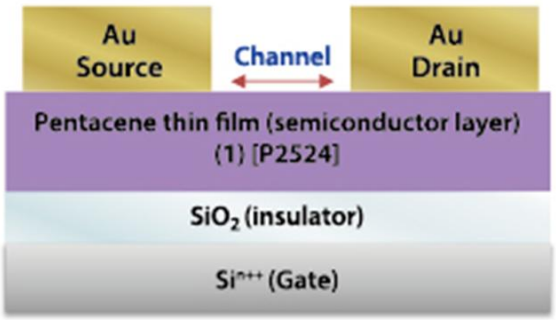
Organic Semiconductor	In-Organic Semiconductor
Composition: Made of carbon-based molecules, polymers, or pi-bonded structures, often including heteroatoms like hydrogen, nitrogen, and oxygen.	**Composition:** Consist of non-carbon-based materials such as silicon, germanium, and gallium arsenide.
Structure & Bonding: Molecules are held together by weak van der Waals forces, forming amorphous or polycrystalline thin films rather than highly ordered crystals.	**Structure & Bonding:** Typically have crystalline structures with strong covalent bonds, forming well-defined energy bands.
Processing & Cost: Can be processed using low-cost, solution-based wet chemistry methods and offer high yields.	**Processing & Cost:** Require more complex manufacturing processes, often involving single-crystal forms.
Flexibility: Inherently flexible and can conform to various shapes.	**Flexibility:** Generally rigid and brittle.
Mobility: Lower electron mobility compared to inorganic semiconductors.	**Mobility:** Exhibit high electron mobility due to their optimal crystalline structure.
Stability: More sensitive to humidity, oxygen, and high temperatures.	**Stability:** More stable and durable in various environmental conditions.
Applications: Used in flexible displays (OLEDs), solar cells, and sensors.	**Applications:** Widely used in transistors, diodes, and integrated circuits for traditional electronics.

Pentacene semiconductor chip

A pentacene semiconductor chip is an organic electronic device using pentacene as the semiconducting layer.

Construction:

Substrate: Usually glass or flexible plastic (PET, PEN)
Gate Electrode: Highly doped silicon
Gate Dielectric Layer: 200-300 nm SiO_2
Pentacene Semiconductor Layer: Thin film (30–100 nm)
Source and Drain Electrodes: Gold (Au)
Encapsulation: Parylene-C polymer



Working Principle:

Pentacene behaves like a p-type organic semiconductor in most cases.

- When voltage is applied to the gate electrode, an electric field is formed across the dielectric.
- This field induces charge carriers (holes) in the pentacene layer, forming a conductive channel between source and drain electrodes.
- The amount of current that flows depend on the gate voltage.
- Because of its π-conjugated structure, pentacene allows efficient charge delocalization, leading to relatively high mobility for an organic material (~1 $cm^2/V \cdot s$).

Advantages of pentacene semiconductor chip:

- Low-Cost Fabrication: Can be processed at low temperatures, often using solution or vacuum deposition techniques.
- Lightweight: Suitable for portable and wearable electronics.
- Flexibility: Can be fabricated on flexible substrates (plastic), useful in wearable electronics and bendable displays.
- Transparency (optional): When used with transparent electrodes (like ITO), pentacene devices can be made semi-transparent.
- Large-Area Applications: Easy to manufacture over large surfaces (e.g., roll-to-roll printing).
- Compatibility with Printing Techniques: Enables low-cost mass production compared to silicon IC fabrication.
- Good Mobility (for organic materials): Among the highest mobilities for organic semiconductors (~0.1–3 $cm^2/V \cdot s$).

Resistive RAM (ReRAM) Materials

Resistive Random Access Memory (ReRAM) is an emerging non-volatile memory technology that stores data by changing the resistance of a material.

Unlike conventional memory devices, ReRAM operates by switching between high resistance state (HRS) and low resistance state (LRS) when an electric voltage is applied. The resistance change occurs due to the formation and rupture of conductive filaments inside metal oxide materials.

Metal oxide nanomaterials such as Titanium dioxide (TiO_2), HfO_2, ZnO, and NiO are widely used in ReRAM devices because of their excellent switching behaviour, stability and nanoscale compatibility. Among them, TiO_2 is one of the most studied materials due to its high dielectric constant, good chemical stability, and ease of synthesis.

Synthesis of TiO_2 nanoparticles by sol-gel method

The sol–gel method is one of the most widely used approaches for preparing nanoparticles. This process is based on the gradual transformation of a liquid solution ("sol") into a solid network ("gel"). The process involves following steps.

1. **Preparation of precursor solution:** It involves mixing of titanium alkoxide precursor (titanium isopropoxide) with ethanol.

2. **Hydrolysis and Condensation:** Deionised water and small amount of HCl is added to precursor solution. The hydrolysis causes precursor to form titanium hydroxide. It then undergoes condensation reaction linking Ti-O-Ti bonds to form a three-dimensional network of TiO_2. This process results in formation of Sol.

$$\mathbf{Ti(OCH(CH_3)_2)_4 + 4\ H_2O \rightarrow Ti(OH)_4 + 4\ (CH_3)_2CHOH}$$
$$\mathbf{Ti(OH)_4 \rightarrow TiO_2 + 2H2O}$$

3. **Gel formation:** As hydrolysis and condensation reaction continue, the sol's viscosity increases leading to the formation of gel. This gel contains three-dimensional network containing trapped solvent molecules.

4. **Drying and calcination:** The gel is dried in hot air oven and further calcinated at higher temperature giving nanoparticles.

Properties:

- TiO_2 exhibits reversible switching between high and low resistance states, which is essential for memory storage.
- High dielectric constant: Provides good charge storage capability.
- Thermal and chemical stability: Maintains performance even at high temperatures.
- Fast switching speed: Allows rapid data writing and erasing.
- Low power consumption: Requires very small operating voltage.

Applications:

- Used in next-generation non-volatile memory devices as an alternative to flash memory in computers and mobile devices.
- Used in artificial intelligence and neuromorphic computing to mimic the behavior of biological synapses.
- Used in Internet of Things (IoT) devices for compact and energy-efficient memory storage.
- Used in embedded memory of microprocessors and integrated circuits.
- Used in high-density data storage systems for large-scale data storage.

Display System

A display system is an electronic arrangement used to present visual information (text, images, or video) to the user in a clear and understandable form. It acts as an interface between electronic devices and human vision, converting electrical signals into visible output.

Liquid Crystals

A distinct state of a matter in which degree of molecular ordering is intermediate between the ordered crystalline state and completely disordered liquid state.

Classification:

Liquid crystals are classified into two types

- Thermotropic liquid crystals
- Lyotropic liquid crystals

1) Thermotropic liquid crystals (TLC): The compounds which exhibit liquid crystal behavior with variation of temperature are called thermotropic liquid crystals.

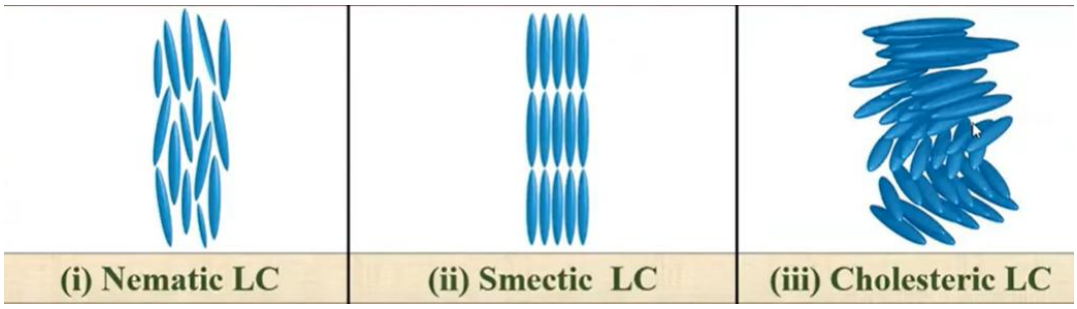
Ex: Cholesteryl Benzoate: (145.5°C & 178.5°C)
P-Azoxy Anisole: (118°C & 135°C)

2) Lyotropic Liquid Crystals: Some of the compounds transformed into liquid crystal phase when mixed with another substance or solvent by the variation of concentration of the compound are called lyotropic liquid crystals

Ex: Soap water mixture
Phospholipid water mixture

Types of Thermotropic liquid crystals:

- **Nematic Liquid Crystals:** These are formed by the compounds that are optically inactive. The molecules oriented along a common axis but are free to rotate around it. They are Used in LCDs
- **Smectic Liquid Crystals:** The molecules form layers with their long axes aligned parallel to one another. They are used in nonlinear optics
- **Cholesteric Liquid Crystals:** The molecules are arranged in a helical structure. They are used in light filters and reflectors.



Properties

- Liquid crystal can flow like a liquid, due to loss of positional order.
- These are elongated and have some degree of rigidity.
- They have less orientational order.
- Transition from crystalline solids to liquid crystals caused by a change of temperature.

Applications

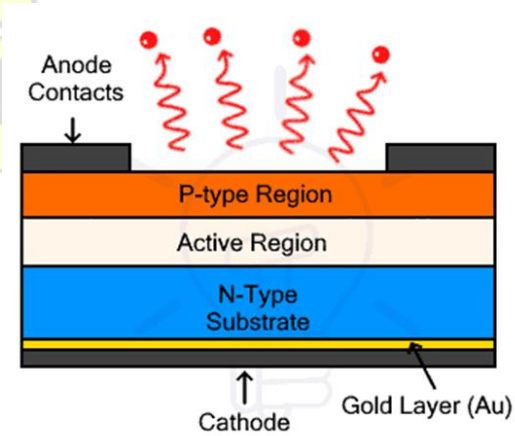
- Liquid crystals are used in watches, calculators, mobile telephones, laptops, computers etc.
- These are used in blood pressure instrument, digital thermometers and TV Chanel indicators.
- These are used in potentiometer, conductometer, Colorimeter etc.

Light Emitting Diodes (LEDs)

An LED display (Light Emitting Diode display) is a type of electronic display that uses light-emitting diodes (LEDs) to produce images, text, and video. It converts electrical signals into visible light to form a display.

Construction:

An LED is a p–n junction semiconductor device that emits light when electrically biased. The construction consists of the following components:



N-type Semiconductor Layer:
Doped with donor impurities → provides electrons (majority carriers).
Common materials:

- Gallium Arsenide (GaAs) – infrared LEDs
- Gallium Phosphide (GaP) – red/green LEDs
- Gallium Nitride (GaN) – blue/white LEDs

P-type Semiconductor Layer: Mg-doped GaN
Anode: Indium Tin Oxide (ITO) (transparent electrode)
Cathode: Gold (Au) or Aluminum (Al)
Active Region: Formed between p-type and n-type layers. This is the region where light is generated.
Encapsulation: Transparent epoxy dome Protects the chip.

Working Principle of LED:

When LED is forward biased:

- Electrons from the n-region and holes from the p-region move towards the junction.
- They recombine at the junction.

- During recombination, energy is released in the form of photons (light) → called electroluminescence.
- The wavelength (color) of light depends on the band gap of the semiconductor material:

  GaAs → Infrared
  GaP → Red/Green
  GaAsP → Yellow/Red

Applications:

- Digital watches, calculators, and clocks.
- TV screens, computer monitors, smartphones.
- Power ON/OFF indicators in electronic gadgets.
- Charging status lights in devices.
- Household and industrial LED lamps.
- Automobile headlights, brake lights, and interior lights.
- Streetlights and traffic signals.

Organic Light Emitting Diode (OLED)

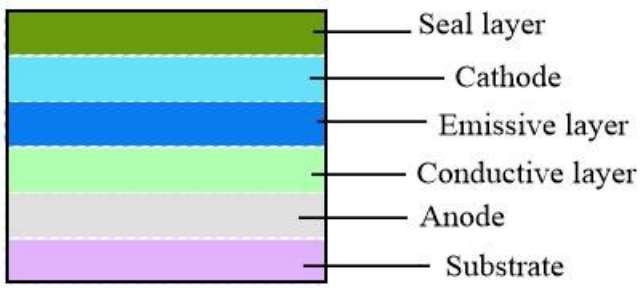
An Organic Light Emitting Diode (OLED) is a thin-film device made of organic semiconductor materials pl aced between two electrodes.

Construction:

Substrate: Usually made of glass or flexible plastic to provide mechanical support.
Anode: Transparent conductive electrode Indium Tin Oxide (ITO).
Organic layers:

- ***Hole Transport Layer (HTL):*** Facilitates the movement of holes from the anode to the emissive layer.
- ***Emissive Layer (EML):*** Organic compound responsible for light emission when electrons and holes recombine.
- ***Electron Transport Layer (ETL):*** Helps electrons move from the cathode to the emissive layer.

Cathode: Metal electrode such as Aluminium or Magnesium.



Working Principle:

- When a voltage is applied between the anode and cathode, current begins to flow.
- The anode injects holes into the hole transport layer and the cathode injects electrons into the electron transport layer.
- Holes move toward the emissive layer through the HTL and electrons move toward the emissive layer through the ETL.
- In the emissive layer, electrons and holes combine to form excitons.
- Excitons release energy as photons, producing visible light.

Applications:

- Used in smartphones and tablets for high contrast and vivid colour displays.
- Used in laptops and monitors for better colour accuracy.
- Used in wearable devices like smartwatches and fitness bands.
- Used in automotive displays, dashboards, and tail lights.
- Used in medical display devices for clear imaging.
- Used in digital signage and advertising panels.

Active Matrix Organic Light Emitting Diodes (AMOLED)

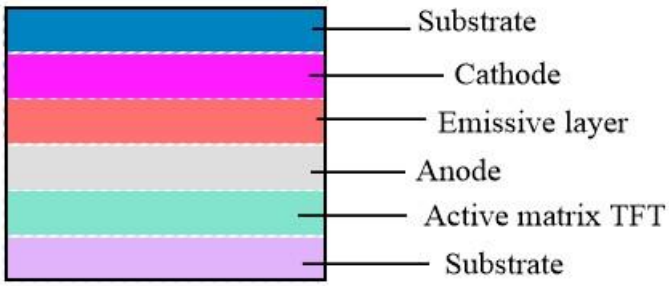
AMOLED is a type of display technology in which organic materials emit light when an electric current is applied, and each pixel is controlled by an active matrix of thin-film transistors.

Construction:

Substrate: Usually, glass or flexible plastic. Provides mechanical support.

Thin Film Transistor (TFT) Layer: Each pixel is connected to a TFT that acts as a switch, allowing precise control of current.

Organic Layers:



- ***Hole Injection Layer (HIL):*** Facilitates injection of holes from the anode.
- ***Hole Transport Layer (HTL):*** Transports holes to the emissive layer.
- ***Emissive Layer (EML):*** Contains organic molecules that emit light when electrons and holes recombine.
- ***Electron Transport Layer (ETL):*** Transports electrons from the cathode to the emissive layer.
- ***Electron Injection Layer (EIL):*** Facilitates electron injection from the cathode.

Anode: Usually transparent ITO.

Cathode: Metals like Calcium or Magnesium.

Working Principle of AMOLED:

- When a voltage is applied to a pixel, the TFT switch activates and allows current to pass through the organic layers.
- Electrons from the cathode and holes from the anode move toward each other and meet in the emissive layer.
- Their recombination releases energy in the form of photons, producing light.
- Each pixel emits light independently, resulting in high contrast, fast response time, and accurate colour control.

Applications of AMOLED:

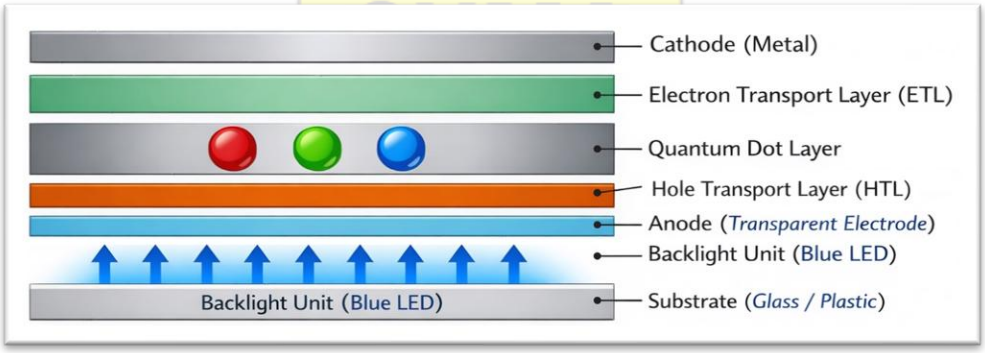
- Used in smartphones for high resolution and low power consumption.
- Used in TVs for high contrast and true black display.
- Used in smartwatches and wearables due to compact size and efficiency.
- Used in tablets and laptops for better viewing angles.
- Used in flexible and foldable devices.
- Used in VR and AR devices for fast response.

Quantum Dot Light Emitting Diode (QLED)

QLED is a display technology that uses quantum dots (semiconductor nanocrystals) to produce bright, pure, and highly accurate colours in display screens.

Construction:

Substrate: Usually, glass or flexible plastic. Provides mechanical support to the device.
Backlight Unit: Typically, blue LEDs are used as a light source.
Anode: Indium Tin Oxide (ITO).
Hole Transport Layer (HTL): Helps movement of holes toward emissive layer.
Quantum Dot (QD) Layer: Contains semiconductor nanocrystals (quantum dots). Converts blue light into pure red and green light with high colour purity.
Electron Transport Layer (ETL): Facilitates electron transport.
Cathode: Metal s like aluminium or calcium.



Working Principle of QLED

- When voltage is applied, the anode injects holes and the cathode injects electrons.
- Electrons and holes move through ETL and HTL respectively toward the quantum dot layer.
- In the emissive layer, electrons recombine with holes.
- This recombination excites quantum dots.
- Excited quantum dots emit light (photons) of specific colors (red, green, or blue).
- In display QLED TVs:
  - Blue backlight passes through QD layer.
  - Quantum dots convert it into pure red and green.
  - Combination produces a wide colour gamut.

Applications of QLED:

- Used in television displays for high brightness and vivid colours.
- Used in computer monitors for improved colour accuracy.
- Applied in smartphones and tablets (emerging technology).
- Used in digital signage and advertising displays.
- Applied in medical imaging displays for precise colour reproduction.
- Used in professional photo and video editing monitors.